

LYCANS AERODESIGN TEAM

Training Task 04

SITL Mission Planning & Geofence Compliance

Deadline: Aug 26, 2026 - 11:59 PM
Submission: GitHub Repository

1 Objective

This task builds directly on Training Session 4 (Flight Control & Autonomy). You will plan and simulate a real competition mission and verify that the aircraft respects its geofence. You will practice:

- Configuring SITL to launch from a custom location
- Building a mission from a provided set of waypoints
- Setting up a geofence and confirming the aircraft stays inside it
- Reasoning about pre-flight preparation and checks for a real flight test

2 Part 1 — Configure SITL at October Airport

By default, SITL launches at a generic simulated location. For this task, you will change that starting location to October Airport.

2.1 What You Need to Do

1. Look up October Airport's coordinates and elevation yourself, and decide on a sensible SITL starting location and heading based on them and on the mission area in Part 2 below.
2. Add your chosen location as a custom SITL location (either via a `--location` argument to `sim_vehicle.py`, or by adding an entry to your local `locations.txt`).
3. Launch SITL from this location and confirm your GCS shows the aircraft starting where you intended.

3 Part 2 — Mission and Geofence Compliance

You will now build the mission using last year's ICMTC competition waypoints and geofence.

3.1 Provided Waypoints

All mission waypoints are listed below in the order they should be flown. Treat every row the same way.

WP #	Latitude	Longitude	Altitude (m)	Radius (m)
1	29.8225135	30.8300829	100	30
2	29.8200934	30.8284521	100	30
3	29.8189252	30.8304584	100	30
4	29.8213779	30.8323359	100	30
5	29.8214687	30.8290181	100	—
6	29.8212918	30.8300427	100	—
7	29.8204844	30.8297691	100	—
8	29.8206403	30.8286747	100	—
9	29.8204471	30.8296537	100	—

3.2 Provided Geofence

Fence Point #	Latitude	Longitude
1	29.8228672	30.8348036
2	29.8249336	30.8305979
3	29.8179897	30.8303511
4	29.8201493	30.8267570

3.3 Requirement

Build the mission exactly as provided above, and configure the polygon geofence to match. Run the mission in SITL from your chosen October Airport starting location.

- The aircraft must strictly follow the defined waypoint path without drifting or deviating significantly from the path line.
- The aircraft must complete the full mission path without ever reaching or breaching the geofence.

4 Part 4 — Pre-Flight Checklist

Simulation success is not the same as being ready to fly a real aircraft. Build your own pre-flight checklist for attempting this exact mission on a real aircraft at a real flight test.

Your checklist should reflect what you actually learned in this task and in Training Session 4. At minimum, it should cover: ground station and mission verification, telemetry and link quality, sensor and calibration status, failsafe and safety checks, and clear go/no-go criteria — but the specific items under each, and any categories beyond these, are for you to work out and justify, not to copy from a reference.

5 Report Requirements

Your submission must be a LaTeX document, using the provided Lycans template. Your report must contain:

- Clear sections and subsections matching Parts 1–3 above.
- A screenshot demonstrating the aircraft actively executing the mission, closely following the waypoint path while staying safely clear of the geofence boundaries.

- The completed waypoint and geofence tables.
- Your written pre-flight scenario and completed checklist from Part 3.

Do not submit screenshots of text or tables when they can be written directly in LaTeX.

6 Tools

For this task, use:

6.1 SITL

Software In The Loop — the same ArduPilot firmware you would fly with, running on your computer instead of real hardware.

6.2 QGroundControl

Use this as your ground station for connecting to SITL, building the mission, and setting the geofence.

7 Submission

Use the same GitHub workflow as previous tasks.

7.1 Task 04 Workflow

1. In your existing personal training repository, create a new branch: `task-04-mission-sitl-geofence`
2. Create a folder: `task-04/`
3. Export your QGroundControl mission plan and save it as `task-04/mission.plan`.
4. Create your assignment document: `task-04/task-04.tex`
5. Complete Parts 1–3 above.
6. Write your report in `task-04/task-04.tex`.
7. Compile and check your PDF using Overleaf (configured for LuaLaTeX engine).
8. Commit your changes (including both `task-04.tex` and `mission.plan`).
9. Push your branch to GitHub.
10. Open a Pull Request from `task-04-mission-sitl-geofence` into `main`.
11. Wait for review.

7.2 Commit Message

Use a clear commit message describing your work.

Example:

```
Add mission and geofence simulation
```



7.3 Pull Request Title

Example:

Task 04 -- SITL Mission & Geofence Compliance

Simulate carefully. Stay inside the fence.